

**PT. INDONESIA POWER**  
**UNIT JASA PEMBANGKITAN PLTU JERANJANG**  
**INTEGRATED MANAGEMENT SYSTEM**  
**28.TE/CBM/UJPJRJ/2020**

|             |                   |        |          |
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**Equipment : Belt Conveyor 10.1**

**KKS : LJ00EBA80AF001-001**

**Technology : Vibrasi dan IRT**

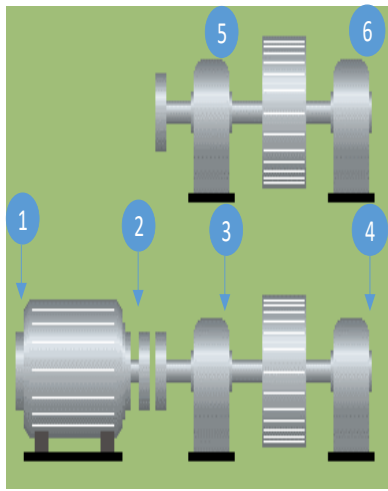
**Tanggal : 5 Oktober 2020**

**Status : Kuning**

**Finding : Vibrasi BC 10.1 Bearing 2 Alarm**

**I. Data Spesifikasi Motor**

| MOTOR            |   | POMPA           |   |
|------------------|---|-----------------|---|
| Type, Mfg        | : | Type, Mfg.      | : |
| Speed            | : | Speed           | : |
| Power            | : | Power           | : |
| Bearing Type     | : | Capacity        | : |
| Inboard Bearing  | : | Type Bearing    | : |
| Outboard Bearing | : | Inboard Bearing | : |
| Rotor Bar        | : | Onboard Bearing | : |
| Foundation       | : |                 |   |

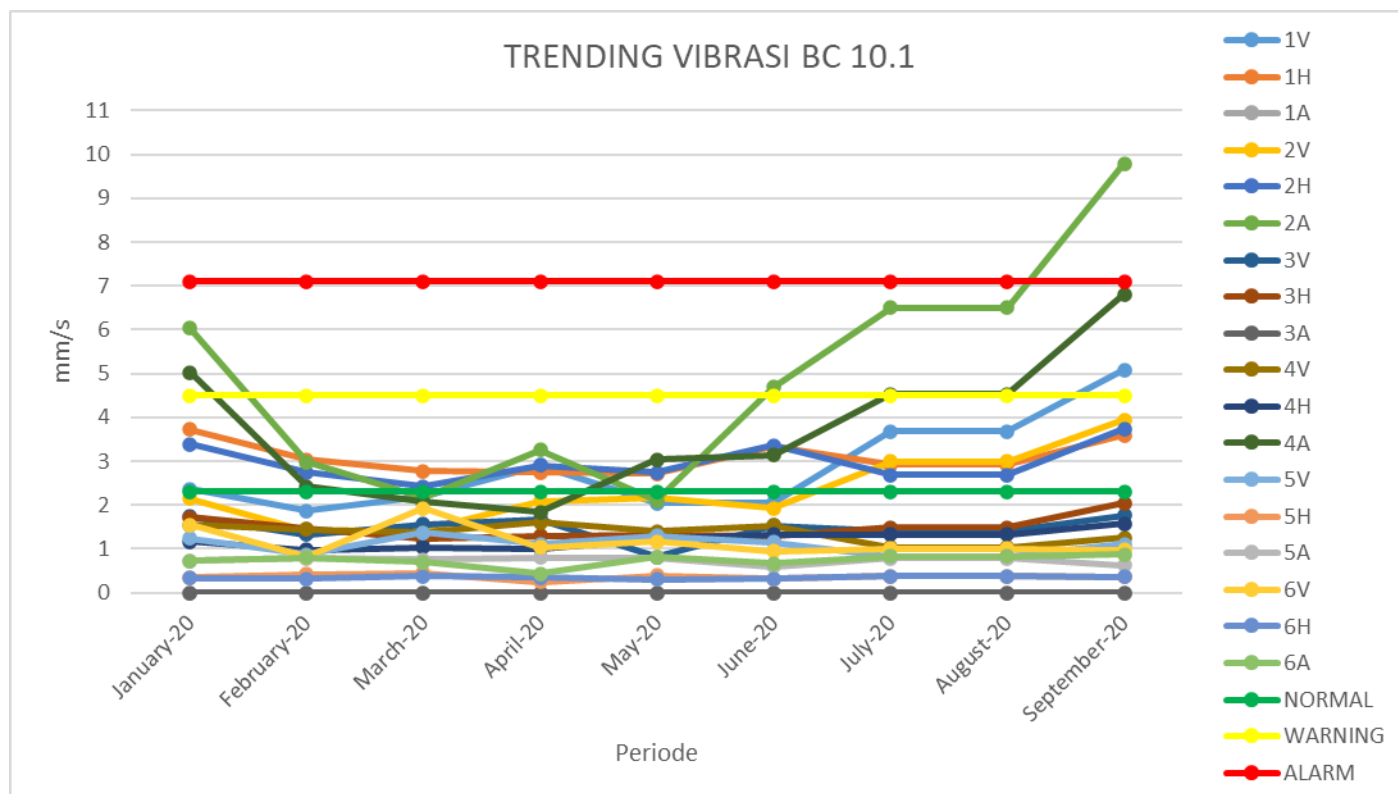
| FOTO EQUIPMENT                                                                      | POINT PENGUKURAN                                                                     |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
|  |  |

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## II. Hasil Pengukuran Vibrasi

| BULAN        | BC 10.1 |      |     |      |      |      |              |      |     |      |      |      |             |      |      |      |      |      |
|--------------|---------|------|-----|------|------|------|--------------|------|-----|------|------|------|-------------|------|------|------|------|------|
|              | MOTOR   |      |     |      |      |      | DRIVE PULLEY |      |     |      |      |      | TAIL PULLEY |      |      |      |      |      |
|              | 1V      | 1H   | 1A  | 2V   | 2H   | 2A   | 3V           | 3H   | 3A  | 4V   | 4H   | 4A   | 5V          | 5H   | 5A   | 6V   | 6H   | 6A   |
| January-20   | 2.36    | 3.72 | N/A | 2.15 | 3.39 | 6.04 | 1.73         | 1.72 | N/A | 1.54 | 1.16 | 5.03 | 1.24        | 0.35 | 0.72 | 1.55 | 0.33 | 0.72 |
| February-20  | 1.86    | 3.04 | N/A | 1.4  | 2.75 | 2.98 | 1.33         | 1.46 | N/A | 1.44 | 0.98 | 2.43 | 0.87        | 0.42 | 0.78 | 0.82 | 0.31 | 0.8  |
| March-20     | 2.21    | 2.77 | N/A | 1.45 | 2.42 | 2.18 | 1.56         | 1.22 | N/A | 1.37 | 1.04 | 2.07 | 1.37        | 0.45 | 0.76 | 1.93 | 0.38 | 0.7  |
| April-20     | 2.91    | 2.75 | N/A | 2.09 | 2.9  | 3.25 | 1.67         | 1.29 | N/A | 1.6  | 1.01 | 1.84 | 1.1         | 0.25 | 0.8  | 1.03 | 0.35 | 0.44 |
| May-20       | 2.05    | 2.72 | N/A | 2.16 | 2.75 | 2.1  | 0.81         | 1.31 | N/A | 1.4  | 1.23 | 3.03 | 1.29        | 0.39 | 0.8  | 1.16 | 0.3  | 0.81 |
| June-20      | 2.06    | 3.33 | N/A | 1.92 | 3.36 | 4.69 | 1.53         | 1.28 | N/A | 1.53 | 1.33 | 3.14 | 1.15        | 0.32 | 0.6  | 0.95 | 0.32 | 0.67 |
| July-20      | 3.68    | 2.92 | N/A | 3    | 2.69 | 6.5  | 1.42         | 1.49 | N/A | 1.04 | 1.32 | 4.53 | 0.82        | 0.38 | 0.78 | 1    | 0.38 | 0.83 |
| August-20    | 3.68    | 2.92 | N/A | 3    | 2.69 | 6.5  | 1.42         | 1.49 | N/A | 1.04 | 1.32 | 4.53 | 0.82        | 0.38 | 0.78 | 1    | 0.38 | 0.83 |
| September-20 | 5.08    | 3.6  | N/A | 3.94 | 3.74 | 9.8  | 1.76         | 2.05 | N/A | 1.26 | 1.57 | 6.81 | 1.1         | 0.36 | 0.63 | 0.97 | 0.36 | 0.88 |



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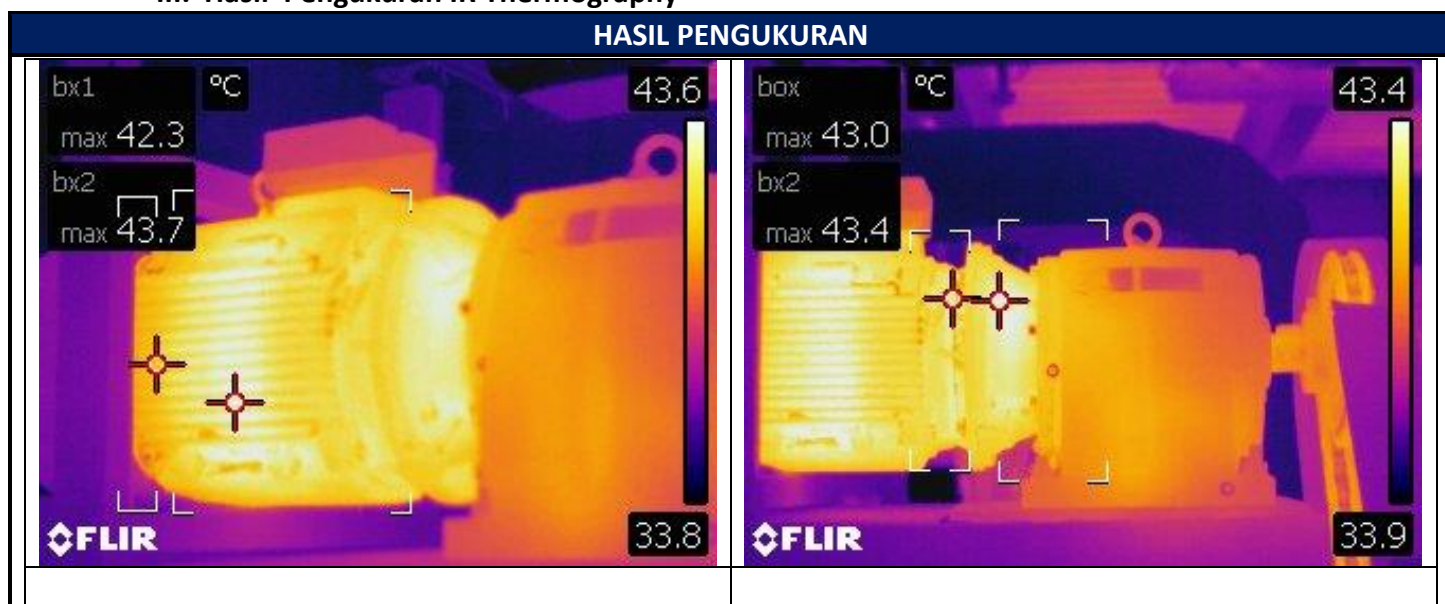
**Hasil Rekaman Spectrum Vibrasi :**

- ✓ **Pada Motor :**
  - Pada Bearing 2 ( NDE Motor) terjadi vibrasi Alarm dengan nilai 9.8 mm/s RMS, Indikasi structure looseness
- ✓ **Pada Drive :**
  - Pada Bearing 4 ( NDE Drive Pulley) terjadi vibrasi Warning dengan nilai 6.81 mm/s RMS, Indikasi structure looseness
- ✓ **Pada Tail Pulley :**
  - Nilai overall Vibrasi Normal

**Rekomendasi :**

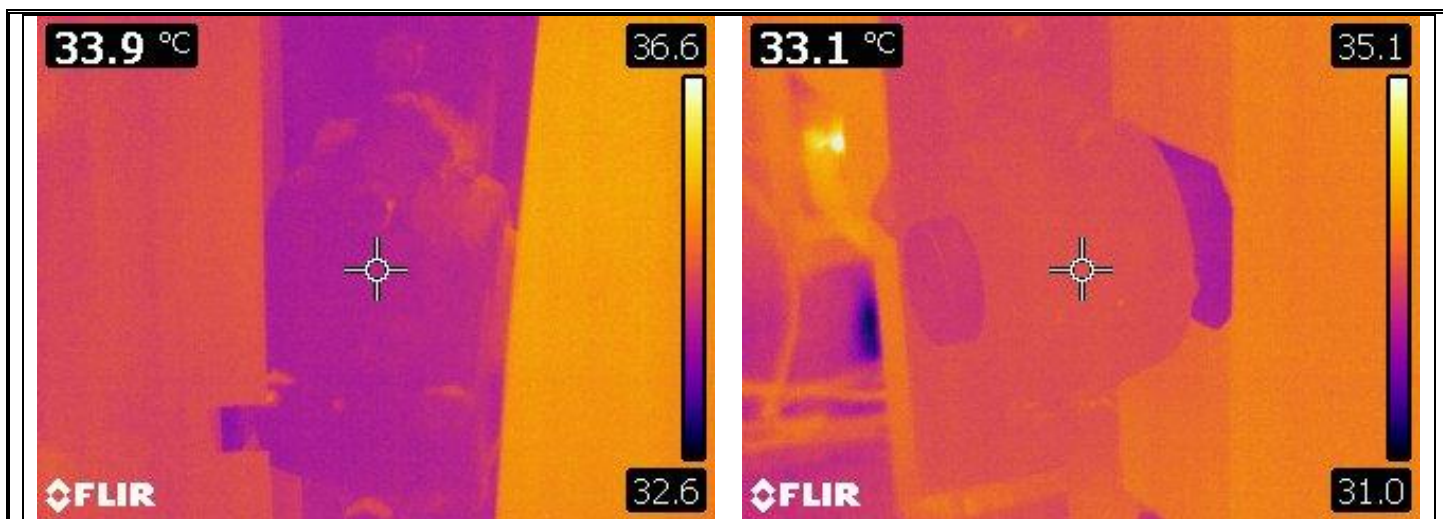
- ✓ **Pada Motor :**
  - Lakukan Pengecekan dan setting kedudukan motor
- ✓ **Pada Drive :**
  - Lakukan Pengecekan dan Setting Rantai
- ✓ **Pada Tail Pulley :**
  - Monitoring

**III. Hasil Pengukuran IR Thermography**



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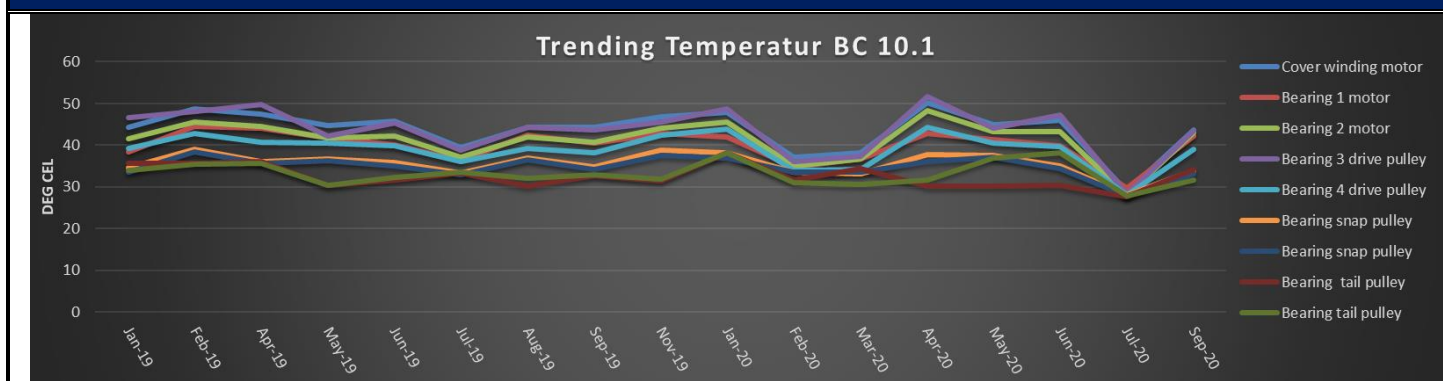
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### NILAI

| Unit | No. KKS            | Description                  | Measurement Point      | Upper Warning Limit Maximo | Upper Action Limit Maximo | No. Point Maximo | Analysis Standard | Status per Point Measurement | Overall Status Equipment | T rise | T max Reference | Data Analysis |        |
|------|--------------------|------------------------------|------------------------|----------------------------|---------------------------|------------------|-------------------|------------------------------|--------------------------|--------|-----------------|---------------|--------|
|      |                    |                              |                        |                            |                           |                  |                   |                              |                          |        |                 | Date          | ( °C ) |
| 0    | LJ00EBA80AF001-001 | EQUIPMENT-BELT CONVEYOR 10.1 | Cover winding motor    | 61                         | 71                        |                  | FLIR              | Normal                       | Normal                   | 0      | 50              | 28-Sep-20     | 43,7   |
|      |                    |                              | Bearing 1 motor        | 51                         | 61                        |                  | FLIR              | Normal                       |                          | 2,3    | 40              | 28-Sep-20     | 42,3   |
|      |                    |                              | Bearing 2 motor        | 56                         | 66                        |                  | FLIR              | Normal                       |                          | 0      | 45              | 28-Sep-20     | 43     |
|      |                    |                              | Bearing 3 drive pulley | 56                         | 66                        |                  | FLIR              | Normal                       |                          | 0      | 45              | 28-Sep-20     | 43,4   |
|      |                    |                              | Bearing 4 drive pulley | 56                         | 66                        |                  | FLIR              | Normal                       |                          | 0      | 45              | 28-Sep-20     | 39     |
|      |                    |                              | Bearing snap pulley    | 41                         | 51                        |                  | FLIR              | Normal                       |                          | 3,9    | 30              | 28-Sep-20     | 33,9   |
|      |                    |                              | Bearing snap pulley    | 41                         | 51                        |                  | FLIR              | Normal                       |                          | 3,1    | 30              | 28-Sep-20     | 33,1   |
|      |                    |                              | Bearing tail pulley    | 41                         | 51                        |                  | FLIR              | Normal                       |                          | 4,1    | 30              | 28-Sep-20     | 34,1   |
|      |                    |                              | Bearing tail pulley    | 41                         | 51                        |                  | FLIR              | Normal                       |                          | 1,7    | 30              | 28-Sep-20     | 31,7   |

### TRENDING

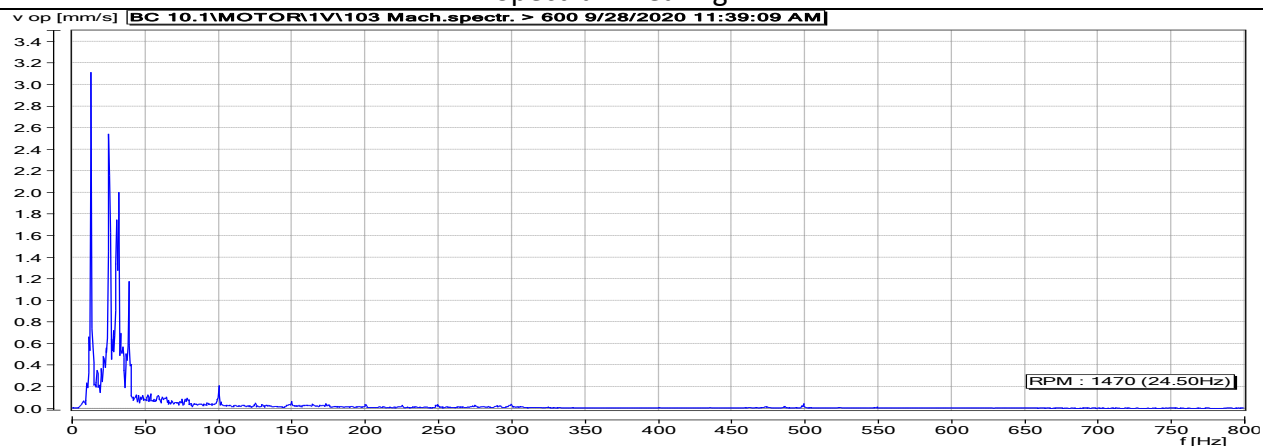


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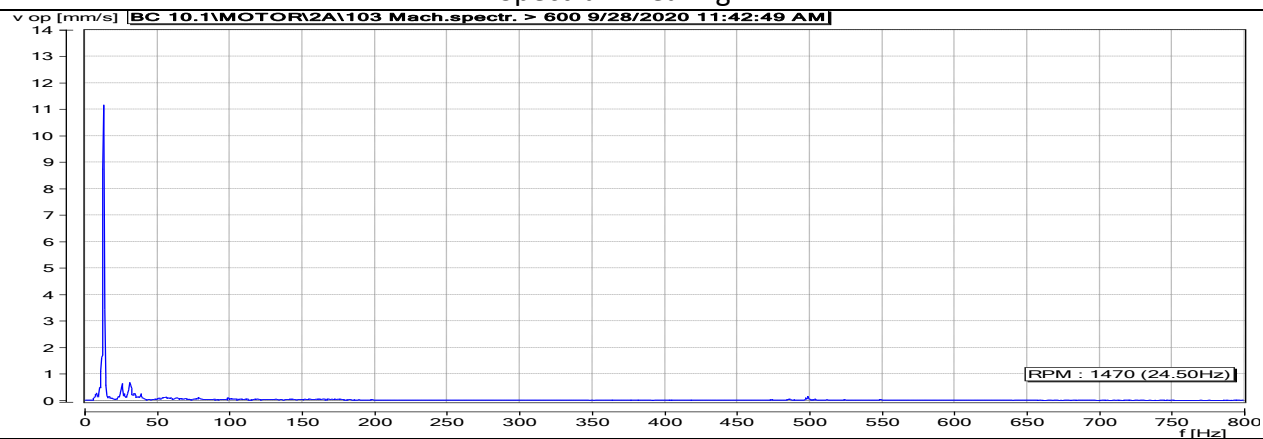
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**IV. Analisa :**

**Spectrum Bearing 1 V**



**Spectrum Bearing 2A**



**Spectrum Bearing 4A**



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**IV. Kesimpulan :**

- Pada sisi motor indikasi terjadi structure looseness karena settinganudukan motor (besi tambahan) berubah atau kurang kencang
- Pada sisi Gearbox dan Drive pully indikasi settingan Rantai coupling kurang pas

**V. Rekomendasi :**

| No | Rekomendasi                                                 | PIC    |
|----|-------------------------------------------------------------|--------|
| 1  | Pengencangan Baut pondasi motor dan baut koneksi semua sisi | HAR EP |
| 2  | Settingudukan tambahan motor                                | HAR EP |
| 3  | Setting Rantai Coupling antara gearbox dan drive pully      | HAR EP |
| 4  | Post Maintenance Test                                       | PdM    |

JERANJANG, 5 OKTOBER 2020

MENGETAHUI  
SPS RSO



RICKY RINALDI

DISUSUN OLEH  
Tim PdM



HERMAWAN